

SPX Fill Realism — August 2026

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INSTRUMENT

SPX (SPXW) — 0DTE cash-settled index options

TRADING HOURS

09:30–16:00 ET (SPXW settles on the close)

MODE

SIMULATED — IB paper fills, checked vs real OPRA
NBBO

TRADING DAYS

22

EXCHANGE

Cboe

BROKER

Interactive Brokers

DATE RANGE

2026-08-06 → 2026-09-04

NBBO / PRINT SOURCE

ThetaData (OPRA consolidated, tick)

In plain English

This checks whether our simulated option fills could really have happened. For each of our **518** buy/sell fills, we looked up the **real market** — the actual bid/ask and the real trades that printed — at the exact second the broker says we filled, and compared our price against it.

Bottom line: the fills look realistic — the same prices a real marketable order would get.

- **Where we filled.** The typical fill landed at position **0.0** between the bid and the ask (0 = the price you pay to cross the spread, 0.5 = the midpoint, 1 = the best possible price). A value near 0 is what a real market order looks like — **86.1%** of fills were at-or-worse than the midpoint, i.e. not too-good-to-be-true.
- **Was the size really there.** **99.8%** of fills had at least one contract actually available at that price — so the fill wasn't a mirage; real depth was sitting there.
- **Did a real trade happen at our price.** Of the **407** fills we could check, **384** had a genuine single-leg trade print at our price (not a multi-leg package price). A fill without one is **not** impossible — see the note under the table.
- **Anything off.** **2.1%** looked slightly better than the market — a sub-second timing artifact (the quote we captured was the last one just before the fill), not an impossible fill. **1** fill(s) had a too-old quote and were set aside.

FILL EVENTS

518

518 with a quote · 0 no logged price

SCORED FILLS

517

quote ok, fresh, price known

MEDIAN FILL POSITION

0.0

0=crossed · 0.5=mid · 1=touch

CROSSED THE SPREAD

86.1%

pos ≤ 0.5 (conservative)

MEDIAN SIZE AT TOUCH

76

SINGLE-LEG PRINT-CONFIRMED

384/407

cond 0/18 traded at our price

contracts resting at our price (≥ 1 :
99.8% · =0: 0.2%)

THROUGH THE BOOK

2.1%

pos > 1 (timing/data noise)

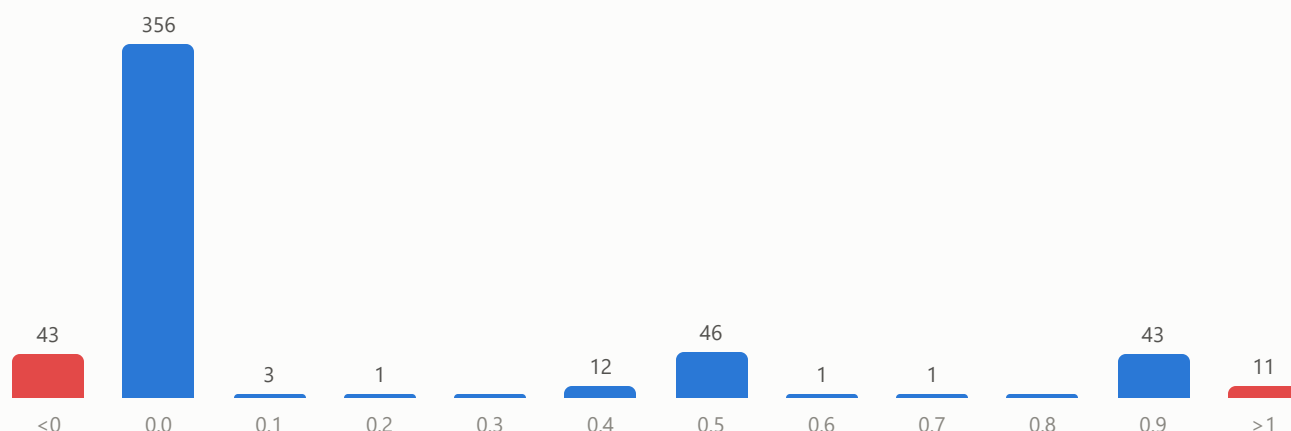
STALE QUOTES

1

quote >2s before fill (excluded)

Where our fills sat in the real NBBO

0 = crossed the spread (marketable, realistic) · 0.5 = mid · 1 = far touch (price improvement) · red = outside the book (timing/data noise). A real marketable engine clusters near 0.



Position breakdown — what share filled where

Every scored fill falls in exactly one band; the shares add to 100%. "Crossed the spread" is the realistic/conservative outcome.

WHERE OUR FILL LANDED	FILLS	% OF FILLS	MEDIAN POSITION
Crossed the spread – filled at the touch	399	77.2%	0.000
Inside – better than the touch, at/below mid	46	8.9%	0.500
Better than mid – price improvement	61	11.8%	1.000
Through the book – sub-second timing noise	11	2.1%	2.000
All scored fills	517	100.0%	0.000

By strategy

scored = quote ok, fresh, our price known. confirm% = single-leg prints (cond 0/18). Strategy names anonymized.

STRATEGY	EVENTS	SCORED	MEDIAN POSITION	CROSSED %	SIZE-OK %	CONFIRM %
S01	74	74	0.000	85%	100%	92%
S02	70	70	0.000	79%	100%	88%
S03	32	32	0.000	94%	100%	100%

STRATEGY	EVENTS	SCORED	MEDIAN POSITION	CROSSED %	SIZE-OK %	CONFIRM %
S04	36	36	0.000	86%	100%	100%
S05	48	47	0.000	96%	100%	95%
S06	44	44	0.000	82%	100%	100%
S07	74	74	0.000	86%	100%	99%
S08	78	78	0.000	79%	99%	89%
S09	22	22	0.000	91%	100%	100%
S10	40	40	0.000	98%	100%	100%

What "print-confirmed" means — and what it doesn't. A single-leg trade printing at our price is *positive* proof the fill was achievable. The reverse is not true: an unconfirmed fill is **not** an impossible one. This is paper, so our own order never prints to the tape — confirmation depends on some *other* trader printing at our exact price within ± 3 seconds. And spread legs frequently trade as multi-leg *package* orders (conditions 130/131/134), which we deliberately exclude. Those fills stand on the NBBO + size-at-touch check instead. In this run, 23 fills had single-leg prints nearby but not at our price, and a further set had no print in the window — neither is a red flag.

Method & FAQ

How each fill is timed — and the lag

Every fill is anchored to **IB's true execution time** (Trade-Confirmation report, second precision). We also record our own fill time in the sim, but that is our machine's clock at the fill *callback*, which fires after the real execution. On the 47 fills we could match, our log lands a median **+1.00s** after IB's execution (mean +2.57s, 90th pct +10.40s; 83% within $\pm 2s$) — our clock+callback offset, which is exactly why we anchor the NBBO lookup on IB's exec time, not our clock.

What we don't yet have: the moment the sim *submitted* the order, so we can't quote a true submit→fill latency. That's one checkbox away — add "Order Time" to the IB Flex query and we compute it exactly.

The ± 3 -second window

The print-confirmation window is **centered on the fill**: 3s before to 3s after (6s total). Why 3s? A trade-off — too narrow ($\pm 1s$) and thin ODTE strikes often have no nearby single-leg print, so we'd under-confirm real fills; too wide ($\pm 9s$) and on a fast tape you match prints from a *different* price regime and over-confirm. $\pm 3s$ catches a genuinely contemporaneous trade without drifting. It affects only the *secondary* print check — the primary metric (where the fill sat in the NBBO) uses the exact quote at the fill instant and is **not** windowed. A volatile 3s is not a problem: more prints just means more chances to find a real trade at our price; we still only count one at/through it.

When would we say a fill could NOT have happened?

Two independent questions:

- **Could it happen? (primary)** At the fill instant, was our price at/inside the real NBBO *and* was size there? We capture the **actual** resting size, not just a flag — median **76 contracts** at our price (range 0–1463; only 0.2% had zero). A fill looks **not** achievable only when it printed **through the book** (better than the best available price — 2.1%) or had **zero size** — and the through-book cases are sub-second quote drift, not impossible fills.
- **Did a real trade print at our price? (secondary)** The 384/407 confirmation. Its absence — 23 had nearby single-leg prints but not at our price, others had none in the window — is **not** evidence against the fill; fillability is judged by the primary test.

What "position < 0" means

Position 0 = filled exactly at the marketable touch (buy at the ask / sell at the bid). **Below 0** means we filled *worse* than the visible touch — paid above the ask or sold below the bid — extra-conservative slippage where the quote moved against us. The **43** fills below 0 make the sim look *harder* on itself, not easier. **Above 1** is the reverse: better than the best price, the through-the-book timing tail.